

Instructions. (15 points) Sections 16.3, 16.4, 16.5. Show relevant work for full credit.

(4^{pts}) 1. Let $\mathbf{F}(x, y, z) = \langle 2xy + z^2, x^2 - 2z, 2xz - 2y \rangle$.

(a) Use the curl test to show that $\mathbf{F}$ is conservative.

(b) Find a potential function $f(x, y, z)$ so that $\nabla f = \mathbf{F}$.

(c) Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ where C is any smooth curve from $(1, 0, 0)$ to $(2, 3, 1)$.

Solution:

(a) With $P = 2xy + z^2$, $Q = x^2 - 2z$, $R = 2xz - 2y$:

$$\nabla \times \mathbf{F} = \langle R_y - Q_z, -(R_x - P_z), Q_x - P_y \rangle = \langle -2 - (-2), -(2z - 2z), 2x - 2x \rangle = \mathbf{0}.$$

Since $\mathbb{R}^3$ is simply connected, $\mathbf{F}$ is conservative.

(b) From $f_x = 2xy + z^2$: $f = x^2y + xz^2 + g(y, z)$. Then $f_y = x^2 + g_y \stackrel{!}{=} x^2 - 2z$, so $g_y = -2z$ and $g = -2yz + h(z)$. Finally $f_z = 2xz - 2y + h'(z) \stackrel{!}{=} 2xz - 2y$, so $h'(z) = 0$. Thus

$$f(x, y, z) = x^2y + xz^2 - 2yz.$$

(c) $f(2, 3, 1) = 12 + 2 - 6 = 8$ and $f(1, 0, 0) = 0$, so $\int_C \mathbf{F} \cdot d\mathbf{r} = 8 - 0 = \boxed{8}$.

(4^{pts}) 2. Use Green's Theorem to evaluate

$$\oint_C (\cos x - y^2) dx + (2xy + y) dy$$

where C is the boundary of the rectangle with vertices $(0, 0)$, $(2, 0)$, $(2, 3)$, $(0, 3)$ traversed counterclockwise.

Solution:

With $P = \cos x - y^2$ and $Q = 2xy + y$: $P_y = -2y$, $Q_x = 2y$, so $Q_x - P_y = 4y$. By Green's Theorem,

$$\oint_C P dx + Q dy = \iint_D 4y dA = \int_0^3 \int_0^2 4y dx dy = \int_0^3 8y dy = 36.$$

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = 36.$$

- (4pts) **3.** Let $\mathbf{F}(x, y) = \langle 2x + y e^{xy}, 3y^2 + x e^{xy} \rangle$, and let C be the curve $y = 2 - 2\cos(\pi x/2)$ from $(0, 0)$ to $(1, 2)$. You may use the fact that $\mathbf{F}$ is conservative on $\mathbb{R}^2$ (no need to verify).

Direct parameterization of C would produce a messy integral. Instead, use path independence to replace C with a two-segment path made of segments parallel to the coordinate axes, and evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$.

Solution:

L-shaped path. The endpoints $(0, 0)$ and $(1, 2)$ are diagonal, so use the two-segment path $C_1 + C_2$ where C_1 : $(0, 0) \rightarrow (1, 0)$ on the x -axis ($y = 0, dy = 0$), and C_2 : $(1, 0) \rightarrow (1, 2)$ on $x = 1$ ($dx = 0$).

Along C_1 : only $P dx$ contributes, and $P(x, 0) = 2x$, so $\int_{C_1} \mathbf{F} \cdot d\mathbf{r} = \int_0^1 2x dx = 1$.

Along C_2 : only $Q dy$ contributes, and $Q(1, y) = 3y^2 + e^y$, so $\int_{C_2} \mathbf{F} \cdot d\mathbf{r} = \int_0^2 (3y^2 + e^y) dy = [y^3 + e^y]_0^2 = 7 + e^2$.

$$\int_C \mathbf{F} \cdot d\mathbf{r} = 1 + (7 + e^2) = \boxed{8 + e^2}.$$

Alternative (potential function). A potential exists: $f(x, y) = x^2 + e^{xy} + y^3$. Check: $f_x = 2x + y e^{xy}$, $f_y = x e^{xy} + 3y^2$. Then by FTLI,

$$\int_C \mathbf{F} \cdot d\mathbf{r} = f(1, 2) - f(0, 0) = (1 + e^2 + 8) - (0 + 1 + 0) = 8 + e^2.$$

- (3pts) **4.** Let $\mathbf{F}(x, y) = \left\langle \frac{-y}{x^2 + y^2}, \frac{x}{x^2 + y^2} \right\rangle$, defined for $(x, y) \neq (0, 0)$. It can be shown that $P_y = Q_x$ everywhere $\mathbf{F}$ is defined. However, if C is the unit circle $x^2 + y^2 = 1$ traversed counterclockwise, it can also be shown that

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = 2\pi \neq 0.$$

The condition $P_y = Q_x$ suggests $\mathbf{F}$ should be conservative, but a nonzero line integral around a closed curve says otherwise. Explain why this is *not* a contradiction of the theorem “if $P_y = Q_x$ then $\mathbf{F}$ is conservative.”

Solution:

The theorem requires $P_y = Q_x$ on a **simply connected** domain (no holes). The domain of $\mathbf{F}$ is the plane with the origin removed, which has a hole at $(0, 0)$ — so it is **not** simply connected. The unit circle C encloses that hole, so Green’s Theorem doesn’t apply on the disk bounded by C (the integrand $Q_x - P_y$ blows up at the origin). The theorem’s hypothesis fails, so there is no contradiction.